

OSCAR CHENG

SOFTWARE ENGINEER | SEEKING INTERNSHIP

CONTACT

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[Personal Website](#)
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EDUCATION

B.S. Computer Science

National Changhua University of
Education (NCUE)
September 2023 – June 2027 (Expected)
Changhua, Taiwan

SKILLS

Languages

Python, C++ (basics), JavaScript,
HTML/CSS

Tools & Workflow

Git / Github, Linux, VS code

Familiar With

LLM API integration, REST APIs, MySQL
basics

LANGUAGES

Mandarin Chinese — Native
English — Intermediate

SUMMARY

Computer Science undergraduate at NCUE specializing in AI systems, LLM applications, and full-stack development. Built end-to-end projects including an AI-powered Market Agent and a RAG-based document analysis system, with experience in FastAPI, vector search, API integration, and deployment. Passionate about building practical AI systems while continuously deepening my understanding of the underlying technologies.

PROJECTS

Market Agent — Trading Signal System

Python, FastAPI, Supabase, OpenRouter, yfinance, scikit-learn • 06 2026

- Built an automated market research system for QQQ / Nasdaq-100 equities, integrating daily price data, technical indicators, news classification, historical signal evidence, and ML-based market references into a structured research report.
- Designed a Supabase-backed data pipeline that updates prices, technical features, market regimes, news events, research outcomes, and freshness checks through scheduled GitHub Actions workflows.
- Developed ML reference models for 5 / 10 / 20-day upside probability, large-drop risk, historical return ranges, and experimental return estimation, with model outputs clearly separated from final investment conclusions.
- Implemented a multi-source evidence quality framework covering data completeness, signal clarity, news coverage, fundamental coverage, backtest sample strength, and future peer / market-wide validation.
- Built a personal frontend dashboard that presents valuation, technical timing, news impact, price plans, ML references, structured data, and data freshness status in a readable research workflow.

RAG Financial Report Analysis

Python, Qdrant, Cross-Encoder, OpenAI API, Streamlit • Fall 2025

- Built a two-stage Retrieval-Augmented Generation pipeline to parse and analyze 100+ page financial reports (Tesla, Palantir), overcoming semantic retrieval ceilings in dense documents.
- Integrated a Cross-Encoder reranking engine (sentence-transformers) and an automated Standalone Query Generator to handle multi-turn context and cross-lingual queries.
- Applied anti-hallucination mechanisms through closed-book prompting, ensuring all generated analyses are grounded in retrieved data.

